

Marcus Kamuntu

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SUMMARY

Third-year Computer Science (Data Science and AI) student at the University of Dundee, graduating 2028. I build and ship production software across the full stack — from LLM inference platforms and agentic RAG pipelines to distributed backend systems and client-facing SaaS products. I founded Sendix (GovTech SaaS, 3 venture awards, 30+ schools waitlist), run a freelance development agency since age 15, and lead the ML engineering component of an industry project with NCR Atleos. I'm most at home when I own a problem end-to-end: from architecture through to production.

EDUCATION

University of Dundee

Scotland, United Kingdom

BSc (Hons) Computer Science (Data Science and AI) with Industrial Placement

2024 – 2028

- Relevant coursework: Data Structures & Algorithms, Multi-Paradigm Programming, Software Engineering, Artificial Intelligence, Information & Network Security
- **Industrial Project (NCR Atleos):** Technical lead on a **Log Aggregation, Analysis & Intelligent Diagnostics Platform** — multi-source ATM log ingestion including real-time Apache Kafka streams, structured querying layer, and ML-powered anomaly detection (scikit-learn Isolation Forest) deployed against production hardware data for a global ATM engineering team.

PROFESSIONAL EXPERIENCE

Sendix

Dundee, Scotland

Founder & Lead Engineer

Jan 2026 – Present

- Founded a **GovTech B2B SaaS** compliance engine automating statutory legal drafting for UK Multi-Academy Trusts — reducing a 3-hour manual process to 15 minutes (**92% reduction**).
- Engineered a **“Glass Box” RAG architecture** on Azure OpenAI with vector database retrieval and deterministic legislative verification, eliminating AI hallucination risk in legally sensitive outputs.
- Built a **Zero-Retention data pipeline** integrating the Wonde API with automated PII tokenisation, redaction, and full audit logging — designed for public-sector GDPR compliance from the ground up.
- Won **three awards** at the University of Dundee Venture 2026 Competition; commercial waitlist of 3 Academy Trusts covering 30+ schools.

Yarn Development

London, England

Founder & Lead Developer

Mar 2021 – Present

- Founded at **age 15** — delivered end-to-end production systems for paying clients across EdTech, hospitality, and professional services, owning requirements, architecture, implementation, CI/CD, and post-launch support.
- Built an EdTech platform (Next.js, Supabase) with AI-generated question banks, RAG feedback sheets, exam builder with synthetic data seeding, and offline-capable print workflows for school environments.
- Managed **client-facing product decisions** throughout — change requests, feature prioritisation, stakeholder communication — developing a product-aware engineering approach from real consequences.

Zamara Uganda

Kampala, Uganda

Pension Funds Intern

Summer 2024

- Contributed to pension fund negotiations with institutional clients including Airtel Uganda and the Makerere University Retirement Benefits Scheme (MURBS) — working directly with senior stakeholders in a regulated financial services environment where accuracy and communication were non-negotiable.

Pollo Game Boat

Boston, MA, USA (Remote)

Co-Founder & Lead Developer

Feb 2021 – Jun 2022

- Built and scaled automated moderation and community tooling serving **250,000+ active users**, migrating backend infrastructure from MySQL to MongoDB under real throughput pressure.
- Developed modular **Node.js/TypeScript APIs** for extensibility and reliability across a globally distributed user base.

PROJECTS

Local-Eval-Stack — Self-Hosted LLM Inference & Evaluation Platform | *Python, llama.cpp, FastAPI, ChromaDB, MLflow, Docker*

- Built an **OpenAI-compatible LLM inference gateway** serving quantised models (Gemma 4 E4B, Llama 3.2 3B) on ARM hardware — FastAPI router with API key auth, per-key rate limiting, and 100% request logging to PostgreSQL.
- **RAG pipeline** with ChromaDB vector store, sentence-transformer embeddings, cross-encoder reranking, and a context budget allocator; paired with an **automated evaluation harness** measuring latency, semantic similarity, and LLM-as-judge quality scores across model variants, tracked in MLflow.
- Wrote a full **Product Requirements Document** covering NFRs, security hardening, fault tolerance design, phased execution plan, and risk register — treating the project as a production engineering initiative from day one.

FluxFile — Distributed File Processing Platform | *TypeScript, BullMQ, Redis, Docker, WebSockets* Feb 2026

- Architected a **distributed async processing system** — containerised worker pool, BullMQ/Redis job queue with backpressure handling, and real-time WebSocket event propagation for audio, video, and document conversion.
- Built with **AES-256 encryption at rest**, zero-knowledge file storage (Cloudflare R2), SSO auth, Stripe billing, and full observability (Sentry, Axiom) — production-grade from day one.

Quartz-Lite — Distributed Task Scheduler | *Java, java.util.concurrent, SQL, JUnit* Oct 2025

- Built a **high-availability job orchestration engine** in Java — custom thread-pool manager using `ExecutorService` and `ThreadPoolExecutor`, configurable pool sizing, graceful shutdown, and thread-safe task submission.
- Designed **SQL-backed job persistence** for fault tolerance across failures; Strategy, Factory, and Observer patterns for full decoupling; JUnit suite targeting concurrency edge cases and race conditions.

Autonomous EV Pathfinding Engine | *Python, NetworkX, GeoPandas, A* Algorithm* Dec 2024

- Engineered a **custom A* search algorithm** over a directed graph of **370,000+ nodes** with multi-variable heuristics balancing geodesic distance, live traffic, and EV energy constraints.
- Built a **geospatial data pipeline** over OS Open Roads shapefiles (GeoPandas) with a real-time Tkinter path visualisation interface.

Nicolaou's Maths — EdTech Platform | *Next.js, Supabase, TypeScript, RAG* Sep 2025

- Full-stack EdTech application for a paying client — AI question bank generation (QGenie), RAG-powered feedback sheets, exam builder with synthetic past-paper seeding, and **react-to-print/docx** for offline school workflows.

Campus Companion | *Next.js, Supabase, TypeScript* Mar 2025

- **Winner, DUCS Quackathon 2025** — first place in a competitive 24-hour hackathon.

SKILLS

Languages: Python, TypeScript/JavaScript, Java, SQL, C, Bash

AI & ML: Azure OpenAI, llama.cpp, RAG architecture, vector databases (ChromaDB), scikit-learn, MLflow, LLM evaluation harnesses, prompt engineering, MCP (Model Context Protocol)

Backend & Databases: FastAPI, Node.js, Express, PostgreSQL, MongoDB, Redis, SQL Server

Distributed Systems: Apache Kafka, BullMQ, async job queues, concurrent programming (`java.util.concurrent`), event-driven architecture

Security & Privacy: AES-256 encryption, PII tokenisation, Zero-Retention pipelines, GDPR-compliant data handling, audit logging

Infrastructure: Docker, GitHub Actions, Microsoft Azure, Cloudflare R2, Vercel, Supabase, Linux

Frontend: React, Next.js, Tailwind CSS, HTML/CSS

Practices: Agile/Scrum, CI/CD, technical documentation, architecture diagrams, PRD authoring, code review, REST APIs